

M3Jar

Phase 4C — Contributor Fund Commitment and Principal Refunds · Final Implementation Report

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Executive Summary

Phase 4C delivers the full contributor fund commitment and principal refund lifecycle for M3Jar. Members can lock committed principal into a jar after the cutoff date (preview !' confirm, idempotent), and request partial or full refunds of their Stripe-only contributions using FIFO lot allocation. Refunds are dispatched asynchronously via the internal dispatcher; Stripe webhook events drive finalization (succeeded), reversal (failed/canceled), or safe failure (unknown status). All 67 new tests pass; all workspaces typecheck clean.

Architecture Decisions (D1–D4, locked)

D1 — Reservation account	REFUND_PENDING ledger account (liability/credit-normal) reserves principal between request and Stripe confirmation. Balance invariant: $\text{contributedPrincipal} = \text{CTRB_REFUNDABLE} + \text{CTRB_COMMITTED} + \text{REFUND_PENDING}$.
D2 — Real tables	refund_requests + refund_allocations (not placeholder stubs). Per-allocation tracking with stable idempotency keys (dripjar-refund:<allocationId>).
D3 — Phase gate	status = 'Saving' AND cutoffDate IS NOT NULL AND today UTC >= cutoffDate. Same check for both commitment and refund routes.
D4 — Partial refund UX	Mobile input pre-filled with maxRefundableCents, editable down. Partial refunds are first-class — remaining principal stays in CTRB_REFUNDABLE.

Database Migrations (0012–0017)

0012	INSERT REFUND_PENDING ledger account
0013	commitment_snapshots + commitment_snapshot_allocations
0014	fund_commitments + commitment_allocations
0015	DROP refund_request_placeholders !' real refund_requests + refund_allocations
0016	Indexes for dispatcher, webhook, snapshot, commitment lookups
0017 (CRITICAL)	Expand ft_transaction_type_check to include refund_reservation, refund_finalization, refund_reversal (was blocking refund FT inserts with 500) All 6 migrations applied to dev DB and registered in lib/db/drizzle/meta/_journal.json.

API Routes

POST /jars/:jarId/commitment/preview	Phase gate !' agreement acceptance gate !' NothingToCommit gate !' FIFO lot snapshot. Returns snapshotToken (10-min TTL).
POST /jars/:jarId/commitment/confirm	Validates snapshotToken freshness and agreement currency !' CTRB_REFUNDABLE DR / CTRB_COMMITTED CR !' idempotent on same token.
GET /jars/:jarId/commitment/mine	Returns member's current fund commitment or null.
GET /jars/:jarId/refunds/preview	Returns maxRefundableCents + FIFO lot breakdown (Stripe-only, uncommitted).
POST /jars/:jarId/refunds	Creates refund request + allocations, posts CTRB_REFUNDABLE DR / REFUND_PENDING CR reservation.
GET /jars/:jarId/refunds/:refundId	Returns refund request + allocations (member sees own; organizer sees all).
POST /internal/dispatch-refunds	Finds pending allocations (stripeRefundId IS NULL), calls stripe.refunds.create with stable idempotency key, updates allocation + request status.

POST /webhooks/stripe refund.updated

POST /webhooks/stripe refund.failed

succeeded !' REFUND_PENDING DR / REFUND_CLR CR (finalization); failed/
canceled !' REFUND_PENDING DR / CTRB_REFUNDABLE CR (reversal).
Same reversal path, allocation providerStatus = 'failed'.

Ledger Accounting (double-entry)

Reservation

DR CTRB_REFUNDABLE / CR REFUND_PENDING (FT type: refund_reservation)

Finalization

DR REFUND_PENDING / CR REFUND_CLR (FT type: refund_finalization)

Reversal

DR REFUND_PENDING / CR CTRB_REFUNDABLE (FT type: refund_reversal)

Commitment

DR CTRB_REFUNDABLE / CR CTRB_COMMITTED (FT type: commit_principal)

Terminal state guard: allocations already in {succeeded, failed, cancelled} are silently skipped by webhook handlers — no double-posting.

Key Shared Helpers

lib/refund-helpers.ts

mapStripeRefundStatus() — Stripe 'canceled' (1 L) !' internal 'cancelled' (2 L's); unknown !' null (no mutation). recomputeRefundRequestStatus(), TERMINAL_ALLOCATION_STATES.

lib/lot-allocation.ts

getRefundableLots() — FIFO Stripe-only lots (uncommitted). allocateFromLots() — greedy allocation from oldest-first lots up to requested amount.

Test Coverage — 67 new tests (all green)

phase4c-commitment.test.ts (29)

Parts A–I: phase gate, agreement gate, preview, confirm, idempotency, stale snapshots, multiple members, financial balance, GET mine.

phase4c-refunds.test.ts (38)

Parts A–M: preview, create, REFUND_PENDING reservation, GET, dispatcher, webhook succeeded/failed/canceled/unknown, terminal guard, balance invariant, mapStripeRefundStatus unit tests.

Key test infrastructure discoveries:

- Commitment: jars auto-create a default agreement on POST /jars !' 'NoActiveAgreement' gate is unreachable in normal flow. Tests updated to expect 'AgreementNotAccepted'. Acceptance uses orderBy(desc(createdAt)) to match getCurrentAgreement().
- Refunds: STRIPE_WEBHOOK_SECRET must be set (or route returns 500 before mocks are reached); INTERNAL_REMINDER_TOKEN must be set (or dispatcher returns 503). Both injected via top-level beforeAll/afterAll.

Mobile UX (Phase 4C)

app/commitment/[jarId].tsx

5-state flow: loading !' preview (summary card, snapshotToken TTL, per-lot breakdown, info box) !' confirming !' success. SnapshotStale !' Alert !' re-fetch.

app/refund/[jarId].tsx

5-state flow: loading !' input (pre-filled max, CurrencyInput editable down, partial badge, FIFO lot list, info box) !' submitting !' success (refund ID shown).

app/jar/[id].tsx (updated)

Lock In + Refund action buttons appear on Overview tab when cutoffReached && phase " {Saving, Commitment}. API enforces the exact gate rules.

Expo Router types auto-regenerated to include /commitment/[jarId] and /refund/[jarId]. Mobile typecheck: clean.

Stripe !" Internal Status Mapping

Stripe uses American spelling 'canceled' (1 L) in webhook payloads; M3Jar's internal schema uses British 'cancelled' (2 L's). mapStripeRefundStatus() handles this conversion. Unknown statuses return null — no state mutation, event marked 'ignored'.

Delivery Checklist

- '6 DB migrations applied (0012–0017), registered in _journal.json
- 'Schema rebuilt: 6 new Drizzle tables
- '9 new API endpoints (commitment preview/confirm/mine, refund preview/create/get, dispatch, 2 webhook cases)
- '67 new API tests — all green (29 commitment + 38 refunds)
- 'All existing Phase 1–4B tests continue to pass
- '2 new mobile screens (commitment, refund) + jar detail updated
- 'All workspaces typecheck clean (lib/db, api-server, mobile)
- 'Code committed to main (52f1eab)

Proposed follow-up tasks: scheduler for dispatch-refunds (Task 4 pattern), member refund status push notifications, organizer refund dashboard.